

1 Solution — GIAM 7.1

Problem 2

1.1 Problem

How many *full houses* are there in Yahtzee? (A full house is a pair together with a three-of-a-kind.)

1.2 Setup

Yahtzee is played with five standard six-sided dice, so each die shows a face value in $\{1, 2, 3, 4, 5, 6\}$. Treat the dice as indistinguishable: a *hand* is the multiset of five face values one sees on the table.

A *full house* is then a hand of the form $\{v, v, v, w, w\}$ — three copies of some value v together with two copies of a different value w . The two values must differ: $v \neq w$. (In particular, five-of-a-kind — a *Yahtzee* — is **not** a full house, since only one value appears.)

1.3 Count

A full house is determined by two independent choices:

1. The value v showing on the triple: $v \in \{1, 2, 3, 4, 5, 6\}$, giving 6 possibilities.
2. The value w showing on the pair: w may be any face value *other than* v , giving 5 possibilities.

By the multiplication principle,

$$\# \text{full houses} = 6 \cdot 5 = \boxed{30}.$$

1.4 Enumeration

Each full house corresponds to an off-diagonal cell of a 6×6 grid indexed by (triple value, pair value):

	pair 1	pair 2	pair 3	pair 4	pair 5	pair 6
triple 1	—	11122	11133	11144	11155	11166
triple 2	22211	—	22233	22244	22255	22266
triple 3	33311	33322	—	33344	33355	33366
triple 4	44411	44422	44433	—	44455	44466
triple 5	55511	55522	55533	55544	—	55566
triple 6	66611	66622	66633	66644	66655	—

Table 1.1

Of the 36 cells, the 6 diagonal cells correspond to $v = w$ and are excluded, leaving $36 - 6 = 30$ full houses — confirming the count.

1.5 Remark — Counting *Rolls* Instead of Hands

The count above treats the dice as indistinguishable, so it counts distinct *hands*. If instead one counts *rolls* of five labeled dice — the natural sample space for probability — each hand arises from multiple rolls. Given the triple value v (6 choices) and the pair value w (5 choices), one must additionally choose *which* three of the five dice show v — there are $\binom{5}{3} = 10$ such choices, and the remaining two dice automatically show w . Hence

$$\# \text{rolls producing a full house} = 6 \cdot 5 \cdot \binom{5}{3} = 6 \cdot 5 \cdot 10 = 300.$$

Since there are $6^5 = 7776$ equally likely rolls of five dice, the probability of a full house on a single throw is

$$\Pr(\text{full house}) = \frac{300}{7776} = \frac{25}{648} \approx 3.86\%,$$

a figure that matches the commonly cited Yahtzee odds.